

Q1: Find the local maximum and minimum values and saddle points of the function.

(a) $f(x, y) = x^2 - xy + y^2 + 9x - 6y + 10$

(b) $f(x, y) = (x^2 + y)e^{y/2}$

a) ① First order derivatives

$$f_x = 2x - y + 9 \quad f_y = -x + 2y - 6$$

② Solve $f_x = 0$ $f_y = 0$

$$f_x = 2x - y + 9 \stackrel{\text{set}}{=} 0 \Rightarrow y = 2x + 9$$

$$\begin{aligned} f_y = -x + 2y - 6 &= -x + 2(2x + 9) - 6 \\ &= -x + 4x + 18 - 6 \\ &= 3x + 12 \end{aligned} \quad \stackrel{\text{set}}{=} 0$$

$$\Rightarrow x = -4$$

$$\text{Then } y = 2(-4) + 9 = 1$$

Critical Point: $(-4, 1)$

③ Second partial derivatives at CP

$$f_{xx} = 2 \quad f_{yy} = 2 \quad f_{xy} = -1$$

$$④ D = f_{xx}f_{yy} - (f_{xy})^2 = (2)(2) - (-1)^2 = 4 - 1 = 3 > 0$$

$$\therefore D > 0 \text{ and } f_{xx} > 0 \Rightarrow \text{local minimum at } (-4, 1)$$

b) ① $f_x = 2xe^{y/2} \quad f_y = e^{y/2}(\frac{1}{2}(x^2 + y) + 1)$

② $f_x = 0 \Rightarrow x = 0$

$$\text{Plug into } f_y = e^{y/2}(\frac{1}{2}y + 1) = 0 \Rightarrow y = -2$$

$$\therefore \text{CP at } (0, -2)$$

$$(3) f_{xx} = 2e^{y/2} \quad f_{yy} = e^{y/2} \left(\frac{1}{2}(x^2 + y) + 1 \right)$$

$$f_{xy} = xe^{y/2}$$

At $(0, -2)$:

$$f_{xx} = 2e^{-1} > 0 \quad f_{yy} = \frac{1}{2}e^{-1} > 0$$

$$f_{xy} = 0$$

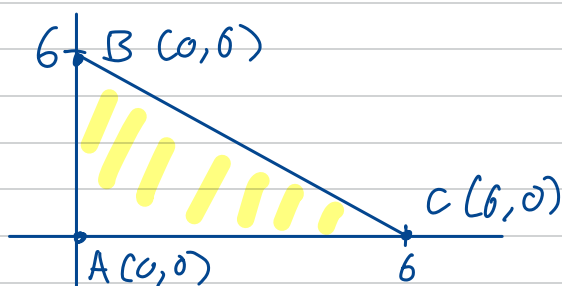
$$(4) D = f_{xx}f_{yy} - (f_{xy})^2 \\ = (+)(+) - 0 > 0$$

$\therefore D > 0$ and $f_{xx} > 0 \Rightarrow$ local minimum at $(0, -2)$

Q3: Find the absolute maximum and minimum values of f on the set D .

$$f(x, y) = 4xy^2 - x^2y^2 - xy^3$$

D is the close triangular region in the xy -plane with vertices $(0,0)$, $(0,6)$, and $(6,0)$.



① Find Critical Points Inside D

$$\begin{cases} f_x = 4y^2 - 2xy^2 - y^3 \\ f_y = 8xy - 2x^2y - 3xy^2 \end{cases}$$

$$\cdot f_x \stackrel{\text{set}}{=} 0 \Rightarrow 4y^2 - 2xy^2 - y^3 = 0 \Rightarrow y^2(4 - 2x - y) = 0 \\ \text{so } y = 0 \text{ or } y = 4 - 2x$$

$$\cdot f_y = 0 \Rightarrow 8xy - 2x^2y - 3xy^2 = 0 \Rightarrow xy(8 - 2x - 3y) = 0 \\ \text{so } x = 0, y = 0, y = (8 - 2x)/3$$

$$4-2x = (8-2x)/3 \Rightarrow x=1 \text{ and } y=4-2(1)=2$$

$\therefore$ CP at $(1,2)$ is inside D . $f(1,2)=4$

(2) Boundary Analysis on D

AB : sub $x=0$ into $f(x,y)$

$$f(0,y) = 0$$

Hence $f=0$ along the whole edge
i.e. It is both the max and min on AB

AC : sub $y=0$ into $f(x,y)$

$$f(x,0) = 0$$

Same here

BC : $y=-x+6$ into $f(x,y)$

$$\Rightarrow f(x, -x+6) = -72x + 24x^2 - 2x^3$$

Now analyze $f(x)$ as in 135:

$$f'(x) = -72 + 48x - 6x^2 \stackrel{\text{set}}{=} 0$$

$$\Rightarrow (x-2)(x-6) = 0$$

$$\Rightarrow \text{CP at } x=2, 6$$

$$f(2) = -72(2) + 24(2)^2 - 2(2)^3 = -64$$

$$f(6) = -432 + 864 - 432 = 0$$

$$f(0) = 0$$

Hence,

Point	$f(x,y)$	
$(0,y)$	0	
$(x,0)$	0	
$(2,4)$	-64	$\leftarrow$ min
$(6,0)$	0	
$(0,6)$	0	

③ Finally, from step ① and ② we get
abs. max: $(1, 2)$ with $f(1, 2) = 4$
abs. min $(2, 4)$ with $f(2, 4) = -64$
inside boundary D

Q4: Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint(s).

(a) $f(x, y) = x^2 y$, $x^2 + y^2 = 1$

(b) $f(x, y) = x e^y$, $x^2 + y^2 = 2$

(c) $f(x, y) = 81x^2 + y^2$, $4x^2 + y^2 = 9$

a) $\nabla f = \lambda \nabla g$

$$\Rightarrow (2xy, x^2) = \lambda (2x, 2y) \quad \text{w/ constraint } x^2 + y^2 = 1$$

$$\Rightarrow \begin{cases} 2xy = 2\lambda x & \Rightarrow xy = \lambda x & (1) \\ x^2 = 2\lambda y & & (2) \end{cases}$$

Case 1: $x=0$

$$\text{Then from constraint } x^2 + y^2 = 1 \Rightarrow y = \pm 1$$

$$\therefore \begin{aligned} (0, 1) & : f = 0 \\ (0, -1) & : f = 0 \end{aligned}$$

Case 2: $x \neq 0$

$$(1): xy = \lambda x \Rightarrow y = \lambda$$

$$\text{sub into (2)} : x^2 = 2\lambda y = 2\lambda^2$$

$$\text{So } x^2 = 2y^2 \Rightarrow \left(\frac{x}{y}\right)^2 = 2 \Rightarrow \frac{x}{y} = \pm \sqrt{2} \Rightarrow x = \pm \sqrt{2}y$$

$$\text{Sub into constraint } x^2 + y^2 = 1$$

$$\Rightarrow 2y^2 + y^2 = 1$$

$$\Rightarrow 3y^2 = 1$$

$$\Rightarrow y = \pm 1/\sqrt{3}$$

$$\text{Then } x = \pm \frac{\sqrt{2}}{\sqrt{3}}$$

So we get 4 CP:

$$\begin{array}{cccc} (1/\sqrt{3}, \sqrt{2}/3) & , & (1/\sqrt{3}, -\sqrt{2}/3) & , & (-1/\sqrt{3}, \sqrt{2}/3) & , & (-1/\sqrt{3}, -\sqrt{2}/3) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ f = 2/3 & & f = 2/3 & & f = -2/3 & & f = -2/3 \end{array}$$

$$c) \nabla f = \lambda \nabla g$$

$$\Rightarrow (162x, 2y) = \lambda (8x, 2y)$$